

## BIRMINGHAM SHUTTLESWORTH, AL, USA

WMO: 722280

Lat: 33.566N

Lon: 86.745W

Elev: 615

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13876

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 20.7       | 24.9 | 6.1                           | 7.7 | 28.4 | 10.3 | 9.5 | 32.1 | 21.1                  | 49.3 | 19.7 | 49.2 | 6.6                      | 0    | 0.400 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.7                   | 95.5            | 74.5 | 93.2 | 74.5 | 91.1 | 74.3 | 78.4                | 88.4 | 77.5 | 87.4 | 76.7 | 86.1 | 7.6                  | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 76.0                            | 138.9 | 82.6 | 74.8 | 133.6 | 81.8 | 73.9 | 129.5 | 81.0 | 42.3          | 88.6 | 41.3 | 87.5 | 40.4 | 86.2 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 18.7              | 16.7 | 14.8 | 14.1                       | 97.8 | 4.6                | 3.4 | 10.8                                               | 100.2 | 8.1        | 102.2 | 5.5        | 104.1 | 2.2        | 106.6 |
|                   |      |      | 12.3                       | 80.1 | 4.4                | 1.6 | 9.2                                                | 81.2  | 6.6        | 82.2  | 4.2        | 83.1  | 1.0        | 84.2  |
|                   |      |      | DB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 63.9   | 44.4  | 48.7  | 55.9  | 63.7  | 71.5  | 78.2  | 81.0  | 80.7  | 75.5  | 64.8  | 54.1  | 47.4 |
|                                                                               | DBStd       | 15.11  | 10.91 | 10.13 | 9.45  | 7.51  | 6.33  | 4.18  | 3.29  | 3.94  | 5.81  | 8.10  | 8.67  | 9.77 |
|                                                                               | HDD50       | 667    | 244   | 136   | 53    | 4     | 0     | 0     | 0     | 0     | 0     | 3     | 56    | 171  |
|                                                                               | HDD65       | 2540   | 641   | 462   | 305   | 112   | 22    | 0     | 0     | 0     | 5     | 105   | 338   | 550  |
|                                                                               | CDD50       | 5740   | 70    | 99    | 234   | 415   | 665   | 845   | 962   | 953   | 765   | 463   | 179   | 90   |
|                                                                               | CDD65       | 2143   | 2     | 5     | 22    | 73    | 223   | 396   | 497   | 488   | 321   | 100   | 12    | 4    |
|                                                                               | CDH74       | 20224  | 1     | 14    | 158   | 605   | 1948  | 3771  | 5093  | 4953  | 2859  | 764   | 54    | 4    |
|                                                                               | CDH80       | 8211   | 0     | 0     | 18    | 107   | 628   | 1593  | 2280  | 2264  | 1131  | 184   | 6     | 0    |
| Wind                                                                          | WSAvg       | 6.0    | 7.1   | 7.3   | 7.5   | 7.2   | 6.0   | 5.3   | 5.0   | 4.6   | 5.0   | 5.3   | 5.7   | 6.5  |
| Precipitation                                                                 | PrecAvg     | 54.5   | 5.1   | 4.6   | 6.1   | 4.8   | 5.0   | 4.2   | 5.1   | 3.9   | 3.9   | 2.8   | 4.3   | 4.8  |
|                                                                               | PrecMax     | 76.5   | 9.3   | 17.7  | 15.8  | 13.7  | 17.2  | 9.1   | 10.2  | 10.9  | 10.4  | 7.5   | 9.6   | 14.0 |
|                                                                               | PrecMin     | 39.2   | 1.1   | 1.2   | 1.7   | 0.4   | 0.9   | 0.7   | 0.3   | 0.4   | 0.2   | 0.1   | 1.7   | 0.8  |
|                                                                               | PrecStd     | 9.0    | 2.1   | 2.9   | 3.1   | 2.9   | 3.2   | 2.3   | 2.5   | 2.6   | 2.8   | 1.8   | 2.0   | 3.0  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 72.3  | 76.3  | 82.6  | 86.1  | 91.5  | 96.4  | 97.4  | 100.0 | 95.8  | 88.3  | 80.1  | 73.4 |
|                                                                               |             | MCWB   | 62.5  | 64.3  | 64.1  | 66.8  | 71.7  | 73.5  | 75.8  | 75.1  | 72.3  | 69.4  | 65.9  | 64.9 |
|                                                                               | 2%          | DB     | 68.4  | 71.8  | 78.2  | 83.1  | 88.4  | 93.1  | 94.7  | 95.9  | 92.5  | 84.5  | 74.7  | 69.9 |
|                                                                               |             | MCWB   | 60.5  | 62.4  | 62.8  | 66.0  | 71.3  | 73.8  | 75.7  | 74.8  | 72.2  | 68.4  | 62.9  | 63.0 |
|                                                                               | 5%          | DB     | 65.0  | 68.4  | 74.9  | 80.5  | 86.2  | 90.7  | 92.4  | 92.9  | 89.5  | 81.5  | 71.7  | 66.7 |
|                                                                               |             | MCWB   | 59.4  | 59.2  | 60.7  | 64.7  | 70.0  | 73.6  | 75.5  | 74.9  | 71.9  | 66.7  | 62.3  | 61.3 |
|                                                                               | 10%         | DB     | 61.8  | 65.1  | 71.4  | 77.6  | 83.7  | 88.4  | 90.1  | 90.2  | 86.6  | 78.4  | 68.6  | 63.2 |
|                                                                               |             | MCWB   | 56.7  | 57.3  | 59.4  | 63.5  | 69.3  | 73.2  | 75.2  | 74.7  | 71.4  | 65.7  | 60.5  | 58.0 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 65.6  | 66.9  | 68.5  | 71.5  | 76.3  | 79.2  | 80.8  | 79.4  | 76.5  | 74.7  | 70.4  | 68.2 |
|                                                                               |             | MCDB   | 69.2  | 72.7  | 74.6  | 79.6  | 85.3  | 89.2  | 90.7  | 91.1  | 85.6  | 80.9  | 73.6  | 71.2 |
|                                                                               | 2%          | WB     | 63.1  | 64.9  | 66.0  | 69.4  | 74.2  | 77.4  | 78.9  | 78.1  | 75.4  | 72.5  | 67.5  | 65.0 |
|                                                                               |             | MCDB   | 66.8  | 69.9  | 73.4  | 77.4  | 83.3  | 87.3  | 88.2  | 88.9  | 84.5  | 80.0  | 70.9  | 68.0 |
|                                                                               | 5%          | WB     | 60.4  | 62.5  | 64.0  | 68.0  | 72.8  | 76.3  | 77.9  | 77.2  | 74.5  | 70.8  | 64.9  | 62.3 |
|                                                                               |             | MCDB   | 64.0  | 66.9  | 71.4  | 76.1  | 81.8  | 85.7  | 87.4  | 87.3  | 83.3  | 77.4  | 69.8  | 66.0 |
|                                                                               | 10%         | WB     | 57.6  | 59.1  | 62.0  | 66.3  | 71.3  | 75.2  | 76.9  | 76.3  | 73.5  | 68.8  | 62.0  | 58.9 |
|                                                                               |             | MCDB   | 61.6  | 63.3  | 69.2  | 74.0  | 80.1  | 84.1  | 86.2  | 85.7  | 81.9  | 74.8  | 66.9  | 62.5 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 18.8  | 19.6  | 21.1  | 21.8  | 19.9  | 18.3  | 17.7  | 18.1  | 19.3  | 20.9  | 20.8  | 18.1 |
|                                                                               |             | MCDBR  | 19.4  | 21.6  | 23.7  | 23.2  | 21.9  | 21.8  | 20.7  | 21.5  | 22.3  | 22.7  | 22.3  | 19.0 |
|                                                                               |             | MCWBR  | 13.6  | 13.4  | 12.1  | 9.9   | 8.0   | 7.3   | 6.4   | 6.4   | 7.3   | 9.5   | 12.5  | 13.8 |
|                                                                               | 5% WB       | MCDBR  | 16.2  | 17.7  | 19.1  | 18.6  | 18.3  | 18.0  | 17.9  | 17.6  | 17.4  | 16.5  | 16.5  | 16.7 |
|                                                                               | MCWBR       | 13.5   | 13.7  | 11.8  | 9.9   | 8.0   | 7.4   | 6.8   | 6.4   | 6.7   | 9.2   | 12.1  | 14.5  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.317  | 0.324 | 0.360 | 0.395 | 0.456 | 0.479 | 0.526 | 0.526 | 0.432 | 0.369 | 0.341 | 0.320 |      |
|                                                                               | taud        | 2.530  | 2.505 | 2.424 | 2.309 | 2.178 | 2.154 | 2.043 | 2.031 | 2.286 | 2.451 | 2.470 | 2.534 |      |
|                                                                               | Ebn at Noon | 286    | 295   | 291   | 284   | 266   | 259   | 246   | 243   | 264   | 274   | 273   | 278   |      |
|                                                                               | Edh at Noon | 26     | 30    | 35    | 41    | 47    | 48    | 54    | 54    | 40    | 31    | 28    | 25    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 831    | 995   | 1362  | 1743  | 1901  | 1951  | 1888  | 1741  | 1528  | 1272  | 939   | 707   |      |
|                                                                               | RadStd      | 63     | 132   | 124   | 125   | 151   | 150   | 121   | 96    | 138   | 164   | 100   | 74    |      |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            |  | +0.83 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -180  | +272  | +194  |
| Regional (40 neighbors) |  | +0.86 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +274  | +198  |

Nomenclature: See separate page